Armand Kassaï Koupaï

PhD student at Sorbonne Université, ISIR, MLIA team

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I am a PhD student at Sorbonne Université working on data-driven methods for solving partial differential equations. My research focuses on neural operators, out-of-distribution generalization, and generative foundation models for scientific applications.

Education

PhD in Computer Science – Sorbonne Université (ISIR) Paris, France
Advisor: Patrick Gallinari *Nov 2022 – Now*
• PhD thesis – Physics-based Deep Learning for parametric PDEs: in and out-of-distribution generalization

MSc in Data Science [w/ distinction]– University of Bristol Bristol, UK
• Machine Learning, Deep learning, NLP, statistics, probability September 2020 – September 2021

Diplôme d'ingénieur (MSc) – ENSEA Cergy-Pontoise, France
• Signal Processing, Computer Science, Applied mathematics September 2018 – September 2021

Selected Publications

Efficient Generative Transformer Operators for Million-Point PDEs **2025**
Armand Kassaï Koupaï, Lise Le Boudec*, Patrick Gallinari* *Preprint*

ENMA: Tokenwise Autoregression for Generative Neural PDE Operators **2025**
Armand Kassaï Koupaï, Lise Le Boudec*, Louis Serrano, Patrick Gallinari* *Spotlight NeurIPS 2025*

Zebra: In-Context and Generative Pretraining for Solving Parametric PDEs **2025**
Louis Serrano, Armand Kassaï Koupaï, Thomas X Wang, Pierre Erbacher, Patrick Gallinari *ICML 2025*

GEPS: Boosting Generalization in Parametric PDE Neural Solvers through Adaptive Conditioning **2024**
Armand Kassaï Koupaï, Jorge Misfut Benet, Yuan Yin, Jean-Noël Vittaut, Patrick Gallinari *NeurIPS 2024*

Learn to adapt parametric solvers under incomplete physics **2024**
Armand Kassaï Koupaï, Yuan Yin, Patrick Gallinari *ICLR Workshop 2024*

Operator learning with neural fields: Tackling pdes on general geometries **2023**
Louis Serrano, Lise Le Boudec, Armand Kassaï Koupaï*, Thomas X Wang, Yuan Yin, Jean-Noël Vittaut, Patrick Gallinari* *NeurIPS 2023*

Self-supervised multimodal fusion transformer for passive activity recognition **2022**
Armand Kassaï Koupaï, Mohammud J Bocus, Raul Santos-Rodriguez, Robert J Piechocki, Ryan McConville *IET WSS*

Experience

University of Bristol
AI Scientist and Researcher in Radar Fusion *Nov 2021– Nov 2022*
- Working for the OPERA project under the supervision of Robert Piechocki, in collaboration with UCL and the University of Oxford.
• *Research on self-supervised learning, multi-modality fusion, transformers*

SustainEcho
Data engineer intern *May 2019 – August 2019*
- The first platform that uses AI to automate the Carbon Footprint of construction projects directly from measurements.
• *Tools: ElasticSearch, Python (NLP), Neo4j*

Renault Group
Signal processing intern *June 2018-August 2018*
- French multinational automobile manufacturer established in 1899. Assisted the R&D team specialized in Sound
• *Tools: Matlab*

Community service

Teaching

Introduction to Data Science (40h) • Bachelor level • Teacher assistant — *Sorbonne Université*
Machine Learning (40h) • Master level • Lecturer — *École Lerebours*
Introduction to Artificial Intelligence (40h) • Bachelor level • Lecturer — *École Lerebours*

Conference reviewer

ICML 2024 - NeurIPS 2024 - ICLR 2025 - AISTATS 2025 – ICML 2025 – NeurIPS 2025 – ICLR 2026

Skills

Programming languages: *Python, Java, C, Shell*
Packages: *Pytorch, Scikit-Learn, Jax, Numpy, Pandas*
Tools: *Linux, Latex, Git, JupyterLab, Weights and Biases*
Languages: *French, English, Persian*